Crankshaft Rear Seal with Retainer Plate

Ford Ecat Electronic Parts Catalog

General Equipment

Plastic Scraper

Material

Item	Specification
Motorcraft® SAE 5W-30 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada)	WSS-M2C946-A
Motorcraft® High Performance Engine RTV Silicone TA-357	WSE-M4G323-A6
Motorcraft® Silicone Gasket Remover ZC-30-A	—
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® Metal Brake Parts Cleaner (US) / Motorcraft® Brake Parts Cleaner (Canada) PM-4-A or PM-4-B (US); CPM-4 (Canada)	—

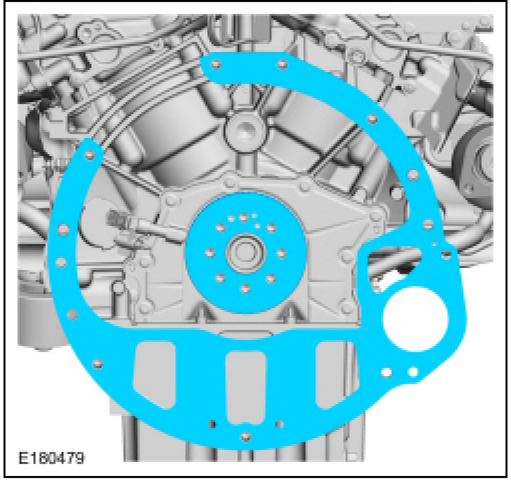
Removal

NOTICE: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces that enters the oil passages, coolant passages or the oil pan, may cause engine failure.

NOTICE: The engine is equipped with a cast aluminum (early build) or stamped steel (late build) crankshaft rear seal retainer plate. All crankshaft rear seal retainer plates must be discarded and a new stamped steel rear seal retainer plate installed.

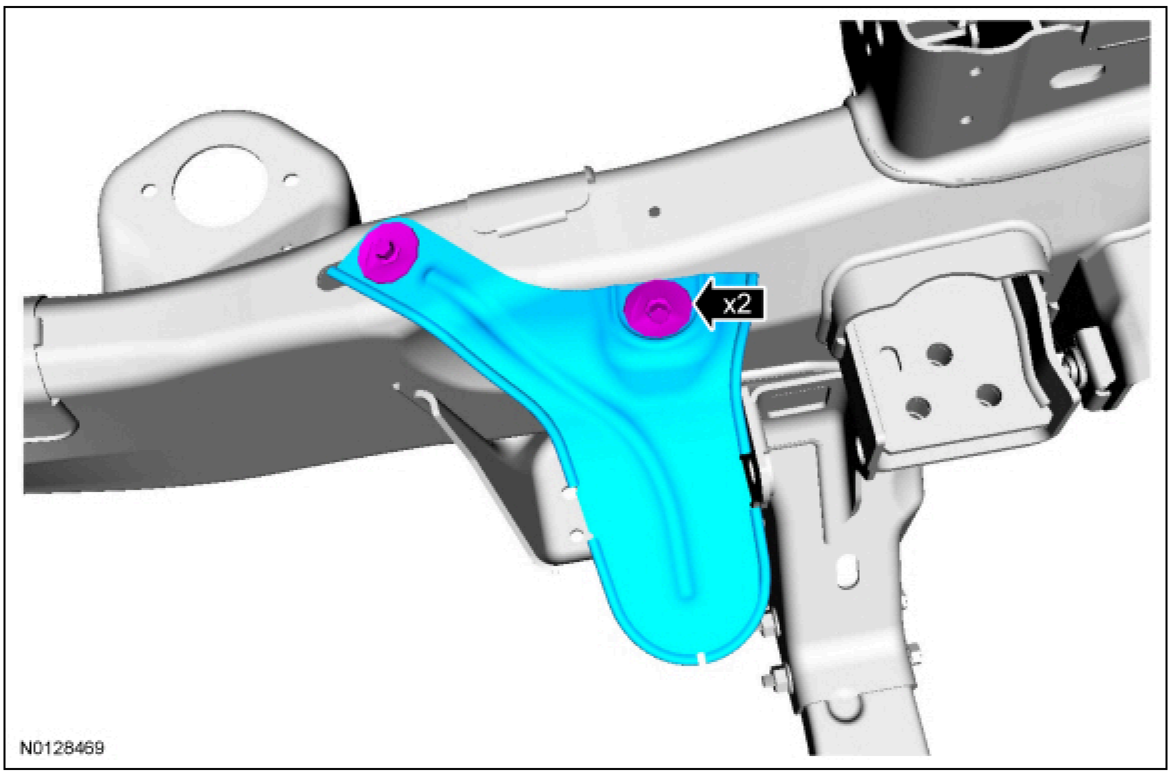
All crankshaft rear seal retainer plates

1. With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
2. Remove the flexplate. Refer to Flexplate in this section.
3. Remove the engine spacer plate and the ignition pulse ring.



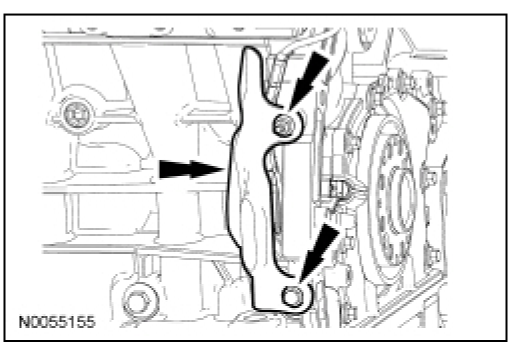
4. If equipped, remove the 4 bolts and the skid plate.

5. If equipped, remove the 2 bolts and the exhaust system heat shield.

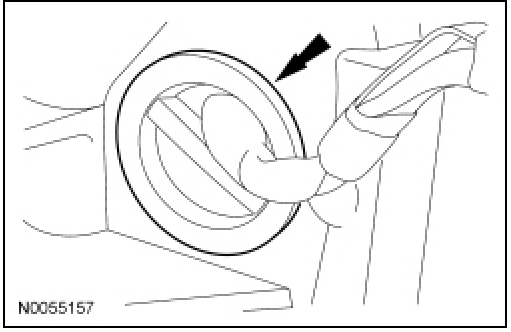


6. **NOTE:** *The Crankshaft Position (CKP) heat shield is only positioned aside.*

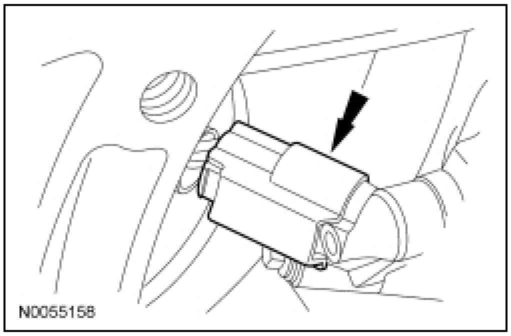
   Remove the nut and the bolt. Position the Crankshaft Position (CKP) heat shield aside.



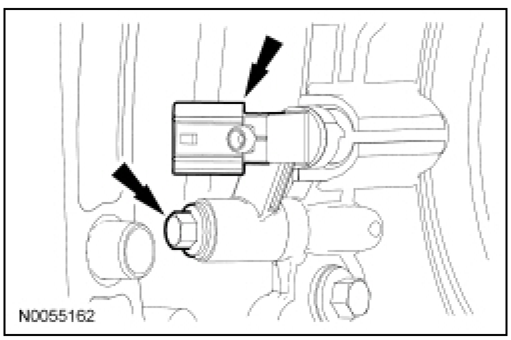
7. Remove the wiring harness grommet.



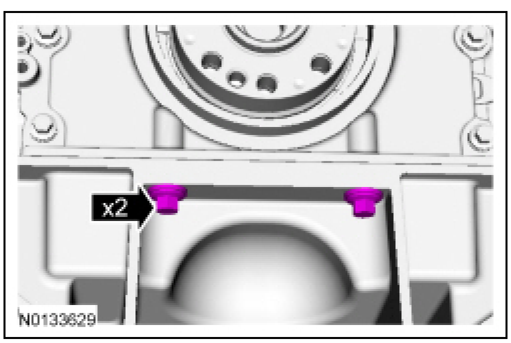
8. Disconnect the CKP sensor electrical connector.



9. Remove the bolt and the CKP sensor.

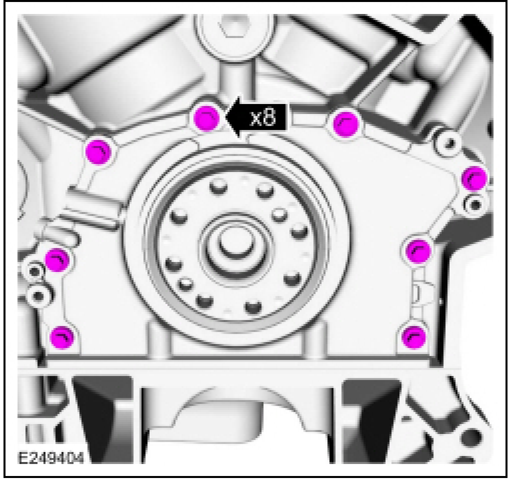


10. Remove the 2 oil pan-to-crankshaft rear seal retainer plate bolts.

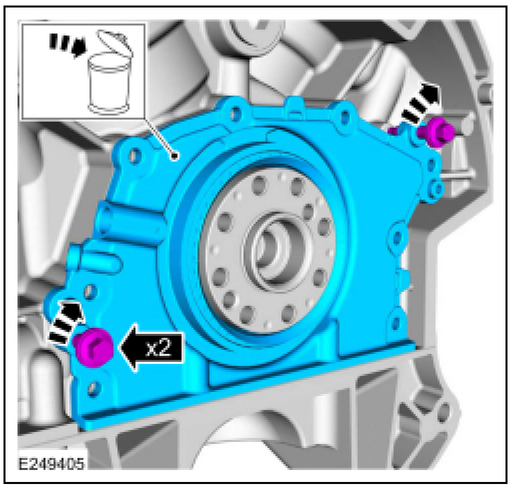


Cast aluminum (early build) crankshaft rear seal retainer plate

11. Remove the bolts.

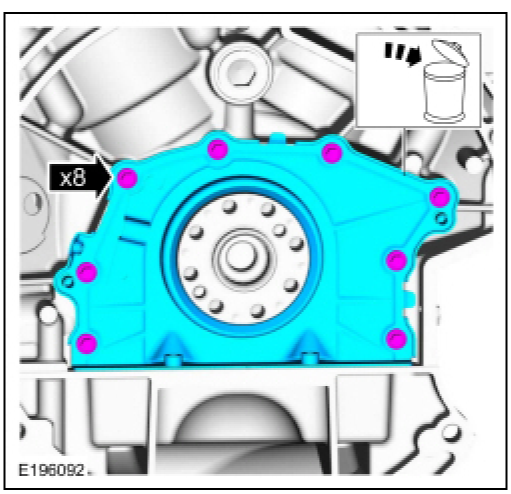


12. Install the 2 M6 oil pan bolts (finger-tight) into the 2 threaded holes in the crankshaft rear seal retainer plate.
    - Alternately tighten the 2 bolts one turn at a time until the crankshaft rear seal retainer plate-to-cylinder block seal is released.
    - Remove the crankshaft rear seal retainer plate.
    - Remove the 2 M6 oil pan bolts.
    - Discard the rear seal and retainer plate.



Stamped steel (late build) crankshaft rear seal retainer plate

13. Remove the bolts and the crankshaft rear seal and retainer plate.
    - Discard the rear seal and retainer plate.



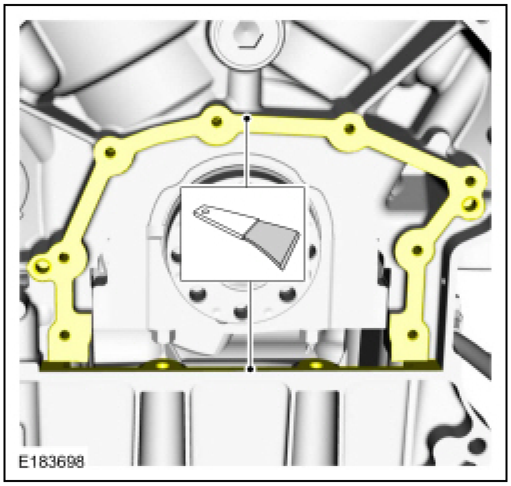
All crankshaft rear seal retainer plates

14. **NOTICE: Place clean, lint-free shop towels over exposed engine cavities. Carefully remove the towels so foreign material is not dropped into the engine. Any foreign material (including any material created while cleaning**

gasket surfaces) that enters the oil passages or the oil pan, may cause engine failure.

NOTICE: Do not use wire brushes, power abrasive discs or 3M Roloc® Bristle Disk (2-in white, part number 07528) to clean the sealing surfaces. These tools cause scratches and gouges that make leak paths. They also cause contamination that will cause premature engine failure. Remove all traces of the gasket.

Make sure that the mating faces are clean and free of foreign material. Refer to Section 303-00.

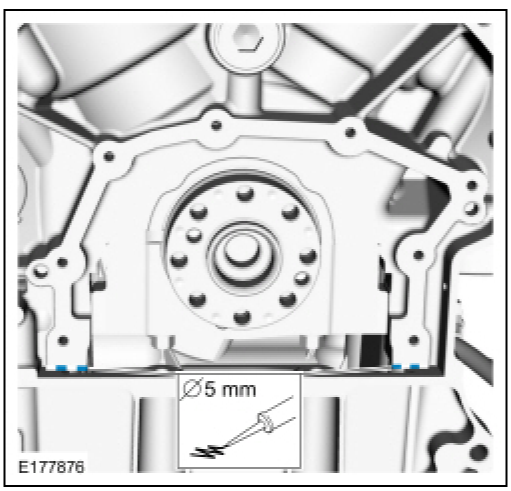


Installation

1. **NOTICE: Failure to use Motorcraft® High Performance Engine RTV Silicone may cause the engine oil to foam excessively and result in serious engine damage.**

   NOTE: *The crankshaft rear seal retainer plate must be installed and the bolts tightened within 10 minutes of sealant application.*

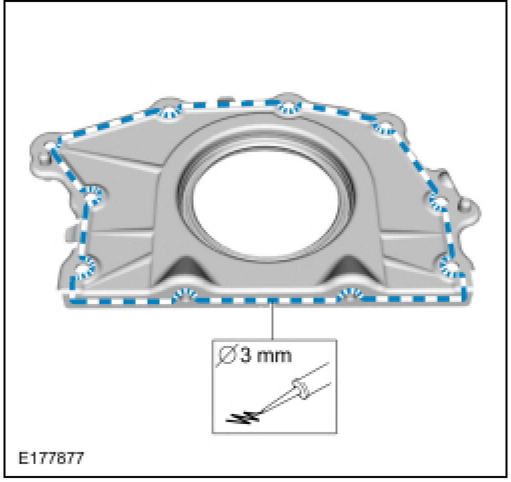
   Apply a 5 mm (0.20 in) bead of Motorcraft® High Performance Engine RTV Silicone to the cylinder block-to-oil pan joints.



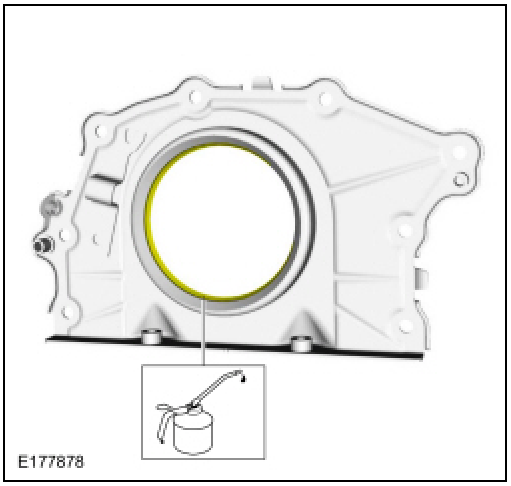
2. **NOTICE: Failure to use Motorcraft® High Performance Engine RTV Silicone may cause the engine oil to foam excessively and result in serious engine damage.**

   NOTE: *The crankshaft rear seal retainer plate must be installed and the bolts tightened within 10 minutes of sealant application.*

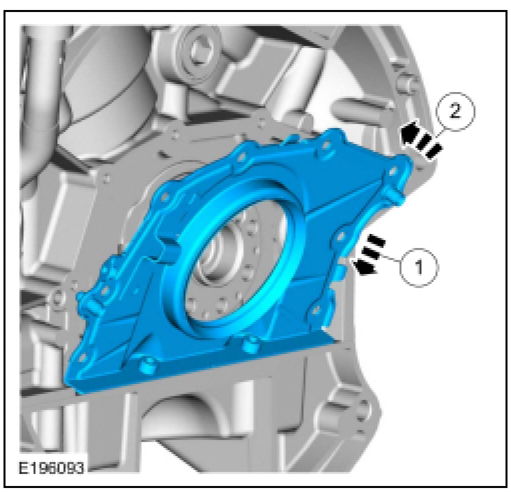
   Apply a 3 mm (0.1181 in) bead of Motorcraft® High Performance Engine RTV Silicone.



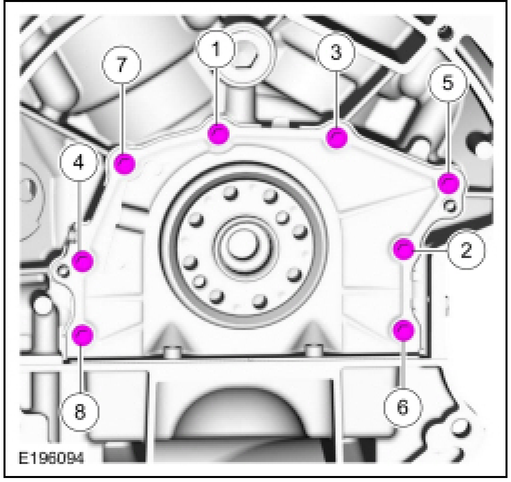
3. Lubricate the crankshaft rear seal with clean engine oil.



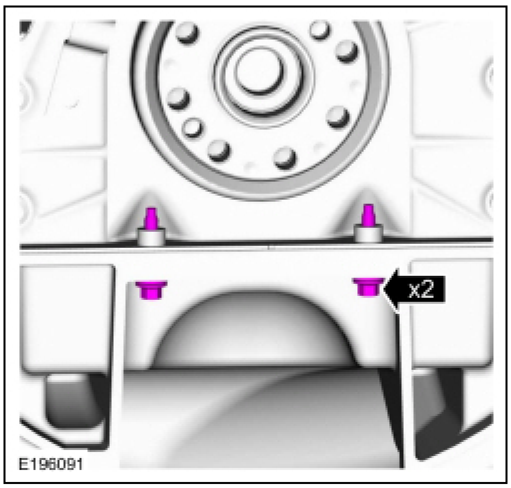
4. Install the crankshaft rear seal retainer plate in the following steps:
    1. Install the crankshaft rear seal retainer plate at an angle above the oil pan flange to avoid scraping off the sealer.
    2. Tilt the crankshaft rear seal retainer plate up and onto the rear of the cylinder block.



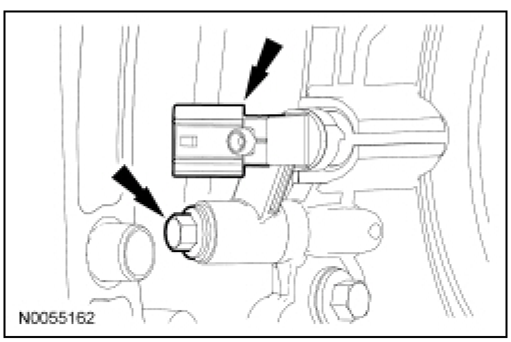
5. Install the bolts and tighten in sequence shown.
    - Tighten to 10 Nm (89 lb-in).



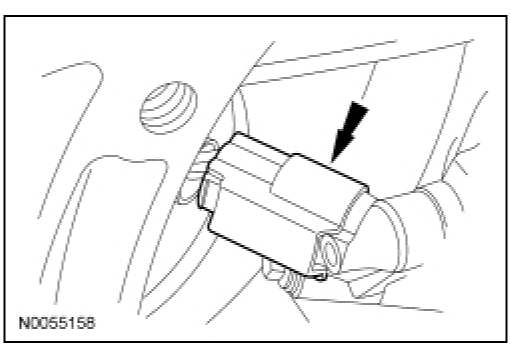
6. Install the oil pan-to-crankshaft rear seal bolts.
    - Tighten to 10 Nm (89 lb-in).



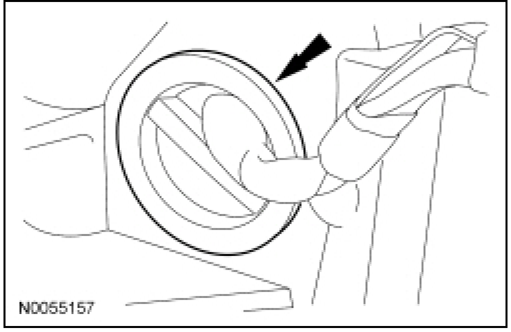
7. Install the CKP sensor and the bolt.
    - Tighten to 10 Nm (89 lb-in).



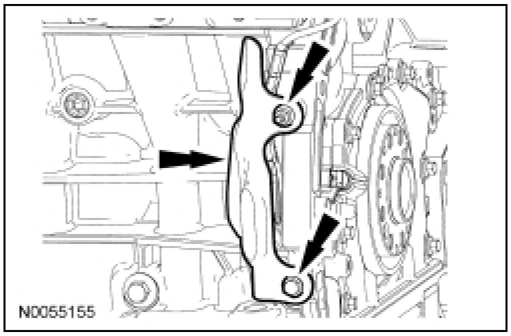
8. Connect the CKP sensor electrical connector.



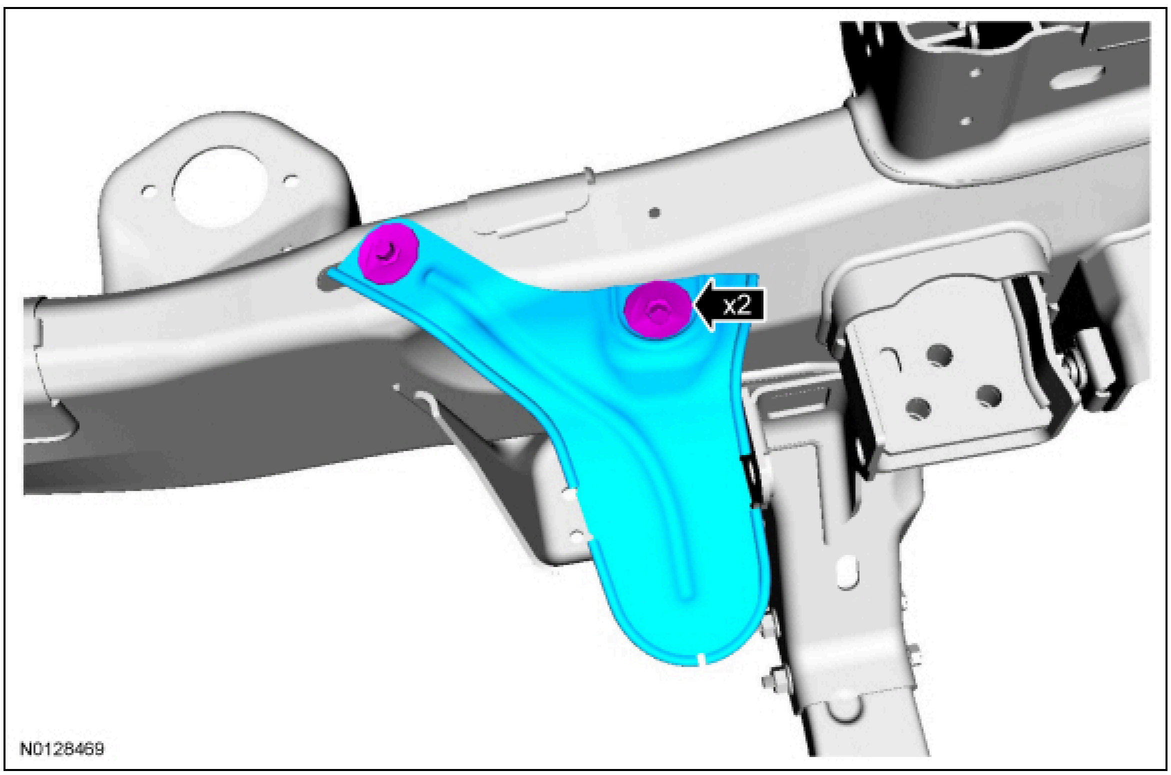
9. Install the wiring harness grommet.



10. Position back the CKP heat shield. Install the nut and bolt.
    - Tighten to 10 Nm (89 lb-in).

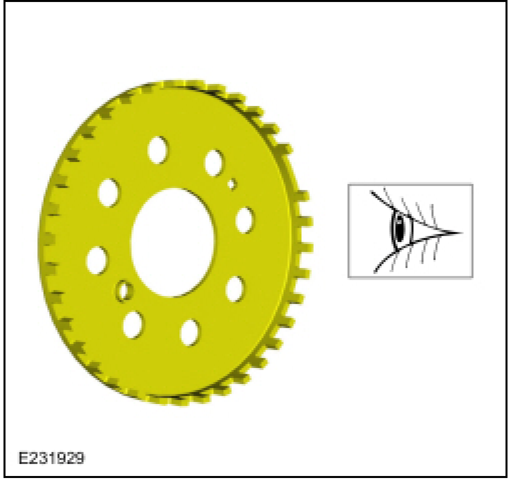


11. If equipped, install the exhaust system heat shield and the 2 bolts.
    - Tighten to 20 Nm (177 lb-in).

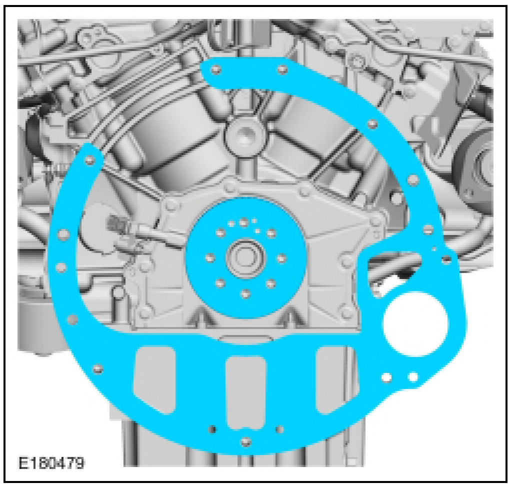


12. If equipped, install the skid plate and the 4 bolts.
    - Tighten to 48 Nm (35 lb-ft).

13. Inspect the ignition pulse ring for damage. If the ignition pulse ring has been dropped or has any visual damage, it must be replaced.



14. Install the engine spacer plate and the ignition pulse ring.



15. Install the flexplate. Refer toFlexplate in this section.